

Real analysis Lecture 3 and 4 homework

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1 Problem 1.11 Let F be an ordered field and $x, y, z \in F$. Then prove if $x < 0$ and $y < z$, then $xy > xz$.

Let $x, y, z \in F$ where $x < 0$ and $y < z$. Since $y < z$ (add $-y$ to both sides) $\Rightarrow 0 = y - y < z - y$ (Additive Inverse). Since $x < 0 \Rightarrow -x > 0$ and $z - y > 0$, therefore $(z - y)(-x) > 0$ (add $(z - y)(x)$ to both sides) $(z - y)(-x) + (z - y)(x) > (z - y)(x) \Rightarrow (z - y)(x + (-x)) = (z - y)(0) = 0 > (z - y)(x) \Rightarrow 0 > xz - yx \Rightarrow yx > xz + (-(yx)) + (yx) = xz$ Thus

$$yx > xz \quad \square$$

2 Let S be an ordered set. Let $A \subset S$ be a nonempty finite subset. Then A is bounded. Furthermore, $\inf A$ exists and is in A and $\sup A$ exists and is in A . Hint: Use induction.

Let S be an ordered subset, let $A \subset S$ be a nonempty finite set. We want to show, A is bounded, $\exists u \in S \mid a \leq u \forall a \in A$ and $\exists l \in S \mid a \geq l \forall a \in A$. Since S is an ordered set and A is a finite set, without the loss of generality let A have n elements and $A = \{x_1, x_2, \dots, x_n\}$ where $x_1 < x_2 < \dots < x_n$. Therefore x_n is an upper bound for A , and A is bounded; also $\forall x \in A \mid x_n \geq x$.

We want to show x_n is the least upper bound. Assume x_n is not the least upper bound for A . There $\exists y \in S \mid \forall x \in A \mid y \geq x$ and $x_n \neq y$. Since S is an ordered set either $y > x_n$ or $y < x_n$ since $x_n \neq y$. If $y > x_n$ then it is not the least upper bound. If $y < x_n \Rightarrow \exists x \in A \mid x > y$ and y is not an upper bound for A . By contradiction x_n is the least upper bound for A . $\perp$

Similarly x_1 is the greatest lower bound for A . $\square$

3 Exercise 1.1.5: Let S be an ordered set. Let $A \subset S$ and suppose that b is an upper bound for A . Suppose that $b \in A$. Show that $b = \sup A$.

Let S be an ordered set. Let $A \subset S$ and suppose that b is an upper bound for A . Suppose that $b \in A$. Since b is an upper bound for A , $\forall x \in A \mid b \geq x$.

Assume b is not the least upper bound for A . There $\exists y \in S \mid \forall x \in A \mid y \geq x$ and $b \neq y$. Since S is an ordered set, either $y > b$ or $y < b$ since $b \neq y$. If $y > b$, then it is not the least upper bound. If $y < b \Rightarrow \exists x \in A \mid x > y$ and y is not an upper bound for A . $\perp$ By contradiction, $b = \sup A$.

4 Let S be an ordered set. Let $A \subset S$ be a nonempty subset that is bounded above. Suppose that $\sup A$ exists and that $\sup A \notin A$. Show that A contains a countably infinite subset. In particular, A is infinite.

Let S be an ordered set. Let $A \subset S$ be a nonempty subset that is bounded above, where $\sup A \notin A$. Assume A is finite, let A have n elements and $A = \{x_1, x_2, \dots, x_n\}$ where $x_1 < x_2 < \dots < x_n$ and has n elements; from problem 2, $x_n = \sup A \Rightarrow x_n \in A$ but $\sup A \notin A$. $\perp$.

By contradiction, A is a non-finite subset $\Rightarrow |A| \geq |N|$. We want to show $|A| \leq |N|$. $\forall x \in A \sup A > x$. The goal is show a sequence exists in A such that $(x_n) \subset A | x_1 < x_2 < \dots < \sup A$.

Base case: Pick any $x_1 \in A$ with $x_1 < \sup A$

Inductive step: Assume we have $x_n \in A$ by the supremum property, there exist x_{n+1} such that

$$x_n < x_{n+1} < \sup A$$

By induction, we can construct $x_1, x_2, \dots$ in A indefinitely. Let $y = \{x_1, x_2, \dots\} \subset A$ and $f: Y \rightarrow N | f(x_n) = n$. Since trivially $a = b \Rightarrow x_a = x_b$. Therefore $|Y| \leq |N| \Rightarrow |A| \leq |N|$. Therefore $|A| = |N|$ $\square$

5 Let A and B be two bounded sets of real numbers. Let $C = \{a + b | a \in A, b \in B\}$. Show that C is a bounded set and that

$$\sup C = \sup A + \sup B$$

and

$$\inf C = \inf A + \inf B$$

Since A and B are bounded sets $\forall a \in A \sup A \geq a$ and $\forall b \in B \sup B \geq b$. Therefore $\forall a \in A, \forall b \in B \sup A + \sup B \geq a + b \Rightarrow \sup A + \sup B = \sup C$. Similarly $\inf C = \inf A + \inf B$. $\square$

6 Let

$$E = \{x \in R : x > 0 \text{ and } x^3 < 2\}.$$

6.1 (a) Prove that E is bounded above.

Let $E = \{x \in R : x > 0 \text{ and } x^3 < 2\}$. Since $x^3 < 2 < 8 \Rightarrow 0 < (8 - x^3) \Rightarrow 0 < (2 - x)(4 + 2x + x^2)$. Since $x > 0$ and therefore $4 + 2x + x^2 > 0 \Rightarrow 0 < 2 - x \Rightarrow x < 2$. Therefore E is bounded above. $\square$

6.2 b) Let $x = \sup E$ (which exists by part (a)). Prove that $x > 0$ and $x^3 = 2$. Hint: Adapt the proof used in Example 1.2.3.

Since $1 > 0$ and $1 < 2, 1 \in E$ and since $x = \sup E \Rightarrow x > 1 \Rightarrow x > 0$ We want to prove $x^3 \geq 2$

Assume $x^3 < 2$, define $h \in E$ $h = \min\{\frac{1}{2}, \frac{2-x^3}{(3x^2+3x+1)}\} < 1$, Since $x^3 < 2 \Rightarrow h > 0$. we will prove $x + h \in E$, we compute $(x + h)^3 = x^3 + 3hx^2 + 3h^2x + h^3 < x^3 + 3h(x^2 + x + 1)$ since $h < 1$. substituting h we get, $x^3 + \frac{(2-x^3)(3x^2+3x+1)}{(3x^2+3x+1)} = x^3 + 2 - x^3 \Rightarrow (x + h)^3 < 2 \Rightarrow x + h \in E$. But this would mean $x + h \leq x \Rightarrow h \leq 0$. By contradiction $x^3 \geq 2$.

We want to show $x^3 \leq 2$

Assume $x^3 > 2$, define $h \in E = \min\{\frac{1}{2}, \frac{x^3-2}{3x^2+1}\} < 1$, note since $2 < x^3 \Rightarrow h > 0$. we will now prove $x - h$ is an upper bound for E . We compute $(x - h)^3 = x^3 - 3x^2h + 3xh^2 - h^3 > x^3 + 3xh^2 - h(3x^2 + 1)$, since $h < 1$. Substituting h , $x^3 + 3x^2h - \frac{(3x^2+1)(x^3-2)}{3x^2+1} = x^3 - x^3 + 2 + xh^2 > 2 \Rightarrow (x - h)^3 > 2$, therefore $(x - h)$ is an upper bound of E . By contradiction $x^3 \leq 2$ Therefore $x^3 = 2$ $\square$